MA 301

Final

Fall - 2014

Problem	points	out of
1		6
2		6
3		6
4		6
5		6
6		6
7		16
8		14
Score		66

Name: ____________________

HR: ____________________

Signature: ____________________

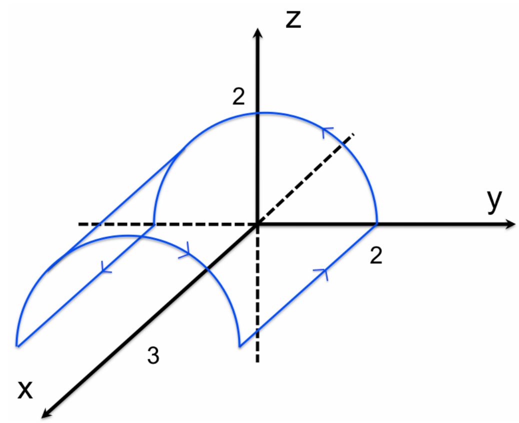
READ BEFORE YOU START
* Show all work.
* Change your calculator mode to RADIANS.
* Sloppy solutions may be penalized.
* Use calculators ONLY for computations.
* You must write up every problem in a clear and organized way.
* If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Let $\mathbf{F} = [16xy + 3yx^2, x^3 + 2y]$ and C is the boundary of the region between: $x = 3$, $y = 0$, & $y = x$.

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

Problem 2: Let $\mathbf{F} = [-y, 2z, 0]$ and C be the boundary of side of the half cylinder $y^2 + z^2 = 4$, with $z \geq 0$, $0 \leq x \leq 3$ and oriented outward (see below).

Use Stokes' thm to compute $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$



Problem 3: Given $\mathbf{F} = [e^x \cos y + yz,\ xz - e^x \sin y,\ xy + z]$

(a) Show that $\mathbf{F}$ is conservative

(b) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is a curve from $(0, 0, 2)$ to $\left(0, \frac{3\pi}{2}, -6\right)$

Problem 4: Let $\mathbf{F} = [x^3 - y^3, y^3 - z^3, z^3 - x^3]$ and let

i. S_S be the upper-half sphere: $x^2 + y^2 + z^2 = 25, \quad z \geq 0$

ii. S_C the region bounded by circle $x^2 + y^2 = 5$ in the x, y-plane.

iii. $S = S_S \cup S_C$

Assuming outward orientation, Compute $\displaystyle\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$

Problem 5: Let $z_1 = -2 + 2i,\ z_2 = -i,\ z_3 = 1 - i.$ Answer the following in the (*specified form*):

(a) (*Standard Form*) $z_1 z_2 =$

(b) (*Standard Form*) $z_3 - z_1 z_2 =$

(c) Convert z_3 to Euler's Polar form:

(d) (*Euler's Polar Form*) $(z_3)^{76} =$

(e) (*Euler's Polar Form*) Solve for x: $\frac{1}{\sqrt{2}} x^3 - z_3 = 0$ (explicitly write out all solutions)

Problem 6: Let $f(x) = \begin{cases} \pi & -\frac{\pi}{2} < x < 0 \\ -\pi & 0 < x < \frac{\pi}{2} \end{cases}$ and $f(x) = f(x+\pi)$

Find the Fourier Series of $f(x)$

Problem 7: Answer the following true/false and short-answer questions:

(a) $\boxed{T}\ \boxed{F}$ If $\mathbf{F}$ is path-independent, then there is a potential function for $\mathbf{F}$

(b) $\boxed{T}\ \boxed{F}$ If $\mathbf{F}$ is path-dependent, then $\int_C \mathbf{F}\cdot d\mathbf{r} \neq 0$ for some closed path C.

(c) $\boxed{T}\ \boxed{F}$ If $\mathbf{F}$ is path-independent, then $\int_C \mathbf{F}\cdot d\mathbf{r} = 0$ for some closed path C.

(d) $\boxed{T}\ \boxed{F}$ Let S be a sphere of radius 1 centered at the origin and oriented outward. If $\iint_S \mathbf{F}\cdot d\mathbf{S} = 0$, then $\mathbf{F} = [0,0,0]$

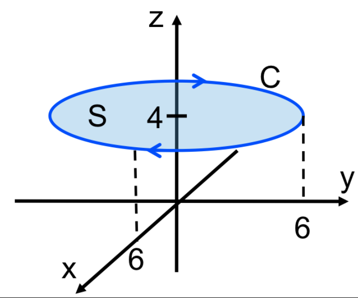
(g) *Fill in the blank:* $\sqrt{-5}\cdot\sqrt{-5} =$ __________

(h) *Fill in the blank:* Suppose $f(x) = f(x+6)$ for all x, then we say f has a period of __________

(e) Let S be the surface of a cube centered at the origin with side length h and outward orientation. If $\mathbf{F} = x\mathbf{i} + xz\mathbf{j} - 3z\mathbf{k}$ and the flux of $\mathbf{F}$ through S is -16. Find h.

(f) If $\mathbf{F} = [xz, 3, y^3]$, $\iint_S \mathbf{F}\cdot\mathbf{n}\, dS = 12\pi$ and S is a disk of radius r in the xz-plane. Find r.

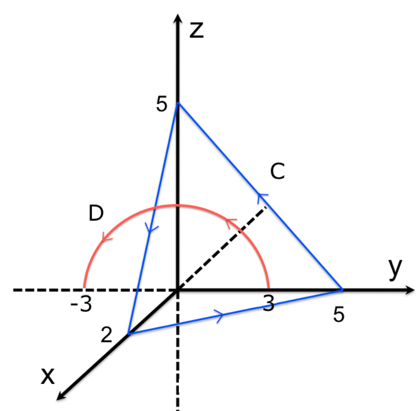
(i) Let $\mathbf{F}(x,y,z) = [zy-3y, xz-z, xy+1]$. Find $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r}$



(j) Let $\mathbf{F} = \nabla\left(\dfrac{y-1}{y+z+1}\right)$ and let C and D be shown in the figure.

$$\int_C \mathbf{F}\cdot d\mathbf{r} =$$

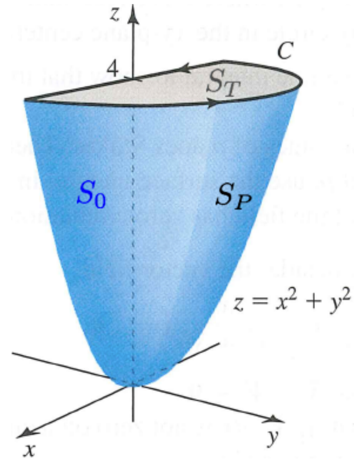
$$\int_D \mathbf{F}\cdot d\mathbf{r} =$$



(k) Let $\mathbf{F} = [6x, ye^z, -e^z]$ and S be the surface (oriented outward) of the box $|x| \le 1, |y| \le 2, 0 \le z \le 1$.

Find $\displaystyle\iint_S \mathbf{F}\cdot\mathbf{n}\, dA =$

Problem 8: Consider the vector field $\mathbf{F}(x, y, z) = [x, y+3, 4+2z]$ and let S_P be part of the paraboloid $z = x^2 + y^2$ (front surface), S_0 (back surface), S_T (top surface), and C (curve around top) be as shown in the figure. Let $S = S_P \cup S_0 \cup S_T$. Finally, let T be the solid bounded by S.



Notes:

1. Assume outward orientation
2. S is closed
3. $\text{Area}(S_0) = 2\displaystyle\int_0^2 \left(4 - x^2\right)\, dx = \frac{32}{3}$
4. $\text{Area}(S_T) = \dfrac{1}{2}\left(\pi(2)^2\right) = 2\pi$
5. $\text{Volume}(T) = 4\pi$

(a) Is $\mathbf{F}$ conservative? Justify your answer.

(b) Which major theorem guarantees that $\displaystyle\iint_{S_P} \text{curl}(\mathbf{F}) \cdot \mathbf{n}\, dS = \iint_{S_T} \text{curl}(\mathbf{F}) \cdot \mathbf{n}\, dS$.

(c) $\displaystyle\iint_{S_T} \text{curl}(\mathbf{F}) \cdot \mathbf{n}\, dS$

(d) Without doing any more work, give $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} =$

(e) $\iint_{S_0} \mathbf{F} \cdot \mathbf{n} \, dS$

(f) $\iint_{S_P} \mathbf{F} \cdot \mathbf{n} \, dS$

(g) $\iint_{S} \mathbf{F} \cdot \mathbf{n} \, dS$

(h) Using (e)-(g), deduce the value of $\iint_{S_T} \mathbf{F} \cdot \mathbf{n} \, dS$